EggGrade IoT Defense guide

Decision: keep the current web model.

A newer model must improve useful results. A higher headline percentage alone is not enough.

Same 649 historical images	Current app	Tested candidate
Correct before recapture	550 / 649 84.75%	572 / 649 88.14%
Correct accepted / all inputs	423 / 649 65.18%	315 / 649 48.54%
Coverage: accepted / all	444 / 649 68.41%	315 / 649 48.54%
Asked for another scan	205	334

Why keep it? The tested candidate returns 108 fewer correct accepted results. Other candidates still miss the Day 16 / Day 18 requirements.

SAY THIS

We compared models using the same inputs. We kept the current model because the candidate increased recaptures and reduced useful correct results.

Read the SAY THIS answer first on each question page. Use the numbers below it if the panel asks for detail.

Tested candidate: grouped-camera-trial-20261019, full-frame RBF classifier. Later development experiments are compared on page 14.

Current release: unfertilized-trial-20261019. Historical results are not field accuracy. No independent 96% or 99% claim is established.

How did you train it?

SAY THIS

We used pretrained image feature extractors. We trained a classifier using labeled egg images, kept each egg in one split, and used validation data to choose settings.

Step	What we did
1. Prepare data	3,719 images. Your 1,590 new views are included. R1, R2 and R3 are excluded.
2. Group eggs	All views of one known egg stay together. We split before training or augmentation.
3. Prepare images	RGB, 224 x 224 pixels, ImageNet normalization. Average original and flipped image features.
4. Train the head	Freeze DINOv2 and EfficientNet-B0. Fit logistic regression on 2,256 training images.
5. Select and evaluate	Use validation for weights and recapture thresholds. Report the 649-image historical evaluation separately.

Training settings: C = 100, seed = 42, per-egg weight = 20. Each egg shares its weight across its views. Class weights balance total effective class weight.

LIMIT / NEXT EVIDENCE

Age labels are user-reported farm ages. Public labels are source annotations. Independent biological verification remains incomplete.

EVIDENCE: train_unfertilized.py; classifier-fit-script-verification.json; manifest and split audit in the technical packet.

Why this split?

SAY THIS

The split protects against seeing the same egg in training and testing. It follows egg identities and source groups, so it is not a fixed 80/20 split.

Partition	Images	D16 / D18 / Reject	Known eggs
Train	2,256 (60.66%)	315 / 327 / 1,614	10
Validation	814 (21.89%)	286 / 290 / 238	10
Historical test	649 (17.45%)	285 / 290 / 74	10

Train learns classifier weights. Validation chooses settings. Historical test reports performance on previously excluded images.

1,590 new photos = 90 three-view photos + 1,500 rotation photos. They represent 30 eggs, not 1,590 independent eggs.

Each split has 530 new photos plus 10 earlier photos. Each split contains five Day 16 and five Day 18 eggs.

There are 2,099 public images: 173 age images and 1,926 Unfertilized images. App name: Reject.

LIMIT / NEXT EVIDENCE

The historical test was inspected during earlier development. It cannot act as a fresh independent confirmation. Public egg identities and batch records are incomplete.

EVIDENCE: split-counts.csv; leakage-audit.json; manifest SHA-256 in model-evidence.json.

Where are the scores?

SAY THIS

Here are precision, recall and F1 for every class. We also show recaptures, because accepted accuracy alone hides how often the app gives no grade.

Class	Images	Precision	Recall	F1
Day 16	285	79.81%	87.37%	83.42%
Day 18	290	86.31%	78.28%	82.10%
Reject*	74	100.00%	100.00%	100.00%

Metric	Result	Meaning
Raw accuracy	84.75%	Correct before recapture gates
Balanced accuracy	88.55%	Average recall across three classes
Macro / weighted F1	88.50% / 84.72%	Equal class weight / image-count weight
Accepted accuracy	95.27%	423 correct out of 444 accepted
Coverage / recapture	68.41% / 31.59%	444 accepted / 205 recaptures
Correct accepted / all	65.18%	423 correct accepted out of 649

Raw accuracy 95% interval: 72.40% to 94.61%. This resamples 66 recorded clusters: 10 eggs and 56 public image families. 5,000 draws; seed 42.

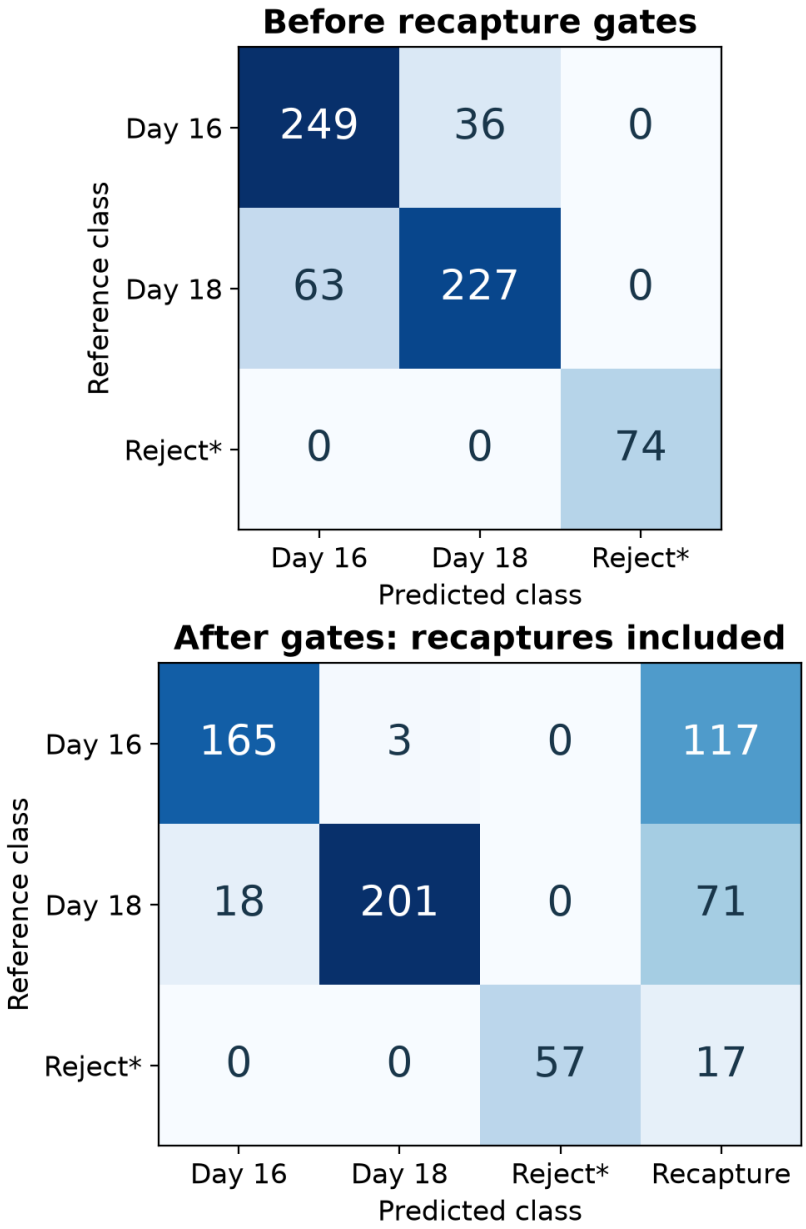
LIMIT / NEXT EVIDENCE

Reject's 100% here means 74 public labeled images were correct. It does not establish 100% performance on real Reject eggs.

EVIDENCE: model-evidence.json; full per-class intervals in classification-metrics.csv. Active model only on this page.

Read rows, then columns

Rows = reference labels. Columns = predictions. Diagonal cells are correct. Other cells are mistakes or recaptures.



Example: 63 Day 18 images were predicted Day 16 before gates. After gates, 18 Day 18 images still received Day 16.

Reject* = public Unfertilized source label. Recapture = no grade issued. They are different outcomes.

EVIDENCE: 649 historical images; unfertilized-trial-20261019. Counts recomputed from saved predictions.

Q4 | COMPUTATION

How did you calculate it?

SAY THIS

We count correct and incorrect predictions in the confusion matrix, then apply the standard formulas. Recaptures stay visible in separate metrics.

For Day 16: TP = 249, FP = 63, FN = 36, TN = 301.

TP: correct Day 16. FP: other class called Day 16. FN: Day 16 called another class. TN: other class called another class.

Metric	Formula	Actual result
Precision	TP / (TP + FP)	249 / 312 = 79.81%
Recall	TP / (TP + FN)	249 / 285 = 87.37%
F1	2TP / (2TP + FP + FN)	498 / 597 = 83.42%
Raw accuracy	Correct / all inputs	550 / 649 = 84.75%
Balanced accuracy	Mean of three recalls	88.55%
Macro F1	Mean of three class F1 scores	88.50%
Weighted F1	Sum(class count x F1) / all	84.72%
Accepted accuracy	Correct accepted / accepted	423 / 444 = 95.27%
Coverage	Accepted / all inputs	444 / 649 = 68.41%
Correct accepted / all	Correct accepted / all inputs	423 / 649 = 65.18%

Zero denominators: class precision, recall or F1 becomes 0; missing support is flagged. Accepted accuracy is undefined if no inputs are accepted.

EVIDENCE: scientific_metrics.py; test_scientific_metrics.py; model-evidence.json. Counts and formulas checked during PDF generation.

Q5 | REPEATABILITY

Does the same egg give the same result?

SAY THIS

Identical saved inputs gave exactly the same scores in our test. Fresh camera scans are a different test. Strict physical zero-gap is not established.

Test	Measured result
A. Same saved inputs	9 three-image bundles, 10 repeats each. 270 fresh view inferences. Model reloaded halfway.
A. Agreement	Same class and accept/recapture decision in every tested bundle.
A. Maximum score difference	0.0 across the complete score vectors.
B. Fresh physical scans	No qualifying study. Score difference not measured.

Saved inputs cover six known age eggs and three public Unfertilized image families. Public family identity is not physical egg identity.

Strict rule: any nonzero full-precision score difference fails. Rounding, reusing predictions or hiding failed scans cannot count as consistency.

LIMIT / NEXT EVIDENCE

ZERO-GAP REQUIREMENT NOT MET. We still need repeated live scans of independently identified eggs from every class.

EVIDENCE: check_saved_repeatability.py; repeatability-results.json; full score logs in the local technical packet.

Why are the ages confused?

SAY THIS

The two ages can look similar under candling. Our model sees appearance, not an incubation clock. Lighting, shell variation and uncertain labels can also affect the result.

Observed before gates	Count
Day 16 predicted Day 18	36 / 285
Day 18 predicted Day 16	63 / 290
Total age errors	99 / 575 age images

Observed in validation: both ages can show a large dark interior and bright lower illumination. Inspected examples pass the quality checks but remain ambiguous.

Biological difference: development continues over the two days. Developmental timing also differs by duck breed. No single brightness cutoff establishes exact age.

Possible causes: light and exposure changes, different shell properties, overlapping appearances, and missing exact incubation records. These are hypotheses, not proven causes of each error.

LIMIT / NEXT EVIDENCE

We need new age-confirmed eggs, documented incubation times and controlled capture conditions to separate these effects.

EVIDENCE: Validation error review and raw matrix; Trela et al., Poultry Science (2025), DOI 10.1016/j.psj.2025.105132; Metzer Farms candling chart.

https://doi.org/10.1016/j.psj.2025.105132
https://www.metzerfarms.com/candling-pictures.html

Which epoch did you stop at?

SAY THIS

The current app uses a logistic classifier, not epoch-based neural training. A separate MLP experiment selected epoch 42 and stopped at 50 in its first inner fold. It was not deployed.

Model	Actual training record
Current app	Frozen encoders + logistic head. Reproduced fit converged in 38 L-BFGS iterations. Maximum 3,000.
Experimental 64-unit MLP	Maximum 60 epochs. Patience 8. First inner fold selected epoch 42; training ended at 50.
Why stop?	Eight epochs gave no improvement under the predefined validation ranking.
Other selected epochs	Five outer refits used 42, 1, 18, 3 and 12 epochs. There is no single best epoch for every fold.

Best checkpoint = selected weights. Final epoch = when training stopped. They need not be the same.

Selection ranks weaker-age recall, mean age recall, public-source F1, then lower loss. Outer evaluation data does not choose the stopping point.

LIMIT / NEXT EVIDENCE

The graph appendix shows the inactive MLP only. Never describe its curves as the web model's training history. Original active solver iterations were not logged; 38 is the verified reproduction.

EVIDENCE: hidden-64/fold-1-inner/history.jsonl and summary.json; train_nested_feature_epochs.py; classifier-fit-script-verification.json.

Did test data leak into training?

SAY THIS

Our checks found no recorded cross-split file, pixel, known-egg or source-family overlaps. That is not proof of zero leakage. Missing public identities and prior test reuse remain limitations.

Check	Result
Recorded file hashes	All 3,719 files match.
Cross-split exact files / pixels	0 overlaps found.
Known eggs / recorded families	0 cross-split overlaps found.
Near-duplicate image check	0 candidates under the declared thresholds.
Development fold groups	0 recorded group crossings.
Learned normalization	Fitted within training folds only.

Near-duplicate rule: 64-bit dHash distance at most 3 AND 32 x 32 RGB mean difference at most 6/255.

LIMIT / NEXT EVIDENCE

Public physical egg identities are unknown. Batch IDs are missing. Reject images come from public sources; camera images cover age classes. The model may learn source cues. The historical test is already inspected.

EVIDENCE: audit_scientific_dataset.py; leakage-audit.json; group folds and source hashes retained in the technical packet.

Which CNN did you use?

SAY THIS

It is a hybrid model. EfficientNet-B0 is the CNN. DINOv2 ViT-S/14 is a transformer. Their frozen features are combined, then a trained logistic head predicts three classes.

Component	Role	Trained here?
EfficientNet-B0	CNN; 1,280 features; ImageNet pretrained	No - frozen
DINOv2 ViT-S/14	Vision transformer; 384 features	No - frozen
Feature fusion	Normalize and combine 1,664 features	No learned fusion
Logistic head	Scores Day 16, Day 18, Unfertilized	Yes
Quality / confidence gates	Request recapture when requirements fail	Selected on validation

The app displays Unfertilized as Reject. This is a source-label mapping, not proof of spoilage, food safety or sellability.

Live workflow: three views. The holder pauses for capture, moves between views, then resumes rotation. Accepted views must agree; at least two accepted views must have distinct hashes.

App model SHA-256 begins 479211e3e587. Full registry, model and camera-head hashes are in model-evidence.json.

LIMIT / NEXT EVIDENCE

A newer experimental architecture is not automatically the model loaded by the web app. Compare the runtime release and hashes.

EVIDENCE: Active model config; model_service.py, including view_consensus and combine_views; authenticated runtime verification.

Q10 | FIELD TEST

What is the field accuracy?

SAY THIS

FIELD ACCURACY NOT YET ESTABLISHED. The reported accuracy comes from saved historical images. We have prepared a separate test using new physical eggs and independent reference labels.

Planned study	Requirement
New eggs	90 total: 30 Day 16, 30 Day 18, 30 confirmed Unfertilized.
Independence	At least three batches per class. Document egg IDs, incubation records, assessor and reference method.
Primary result	First three-view attempt for each of the 90 eggs. Include failed and recapture attempts.
Repeat study	Three eggs per class, ten scans each within ten minutes. 171 total planned attempts, still only 90 eggs.
Success criteria	Raw accuracy and each class recall at least 96%. Coverage at least 68.41%; correct accepted / all at least 65.18%. No paired baseline regression.

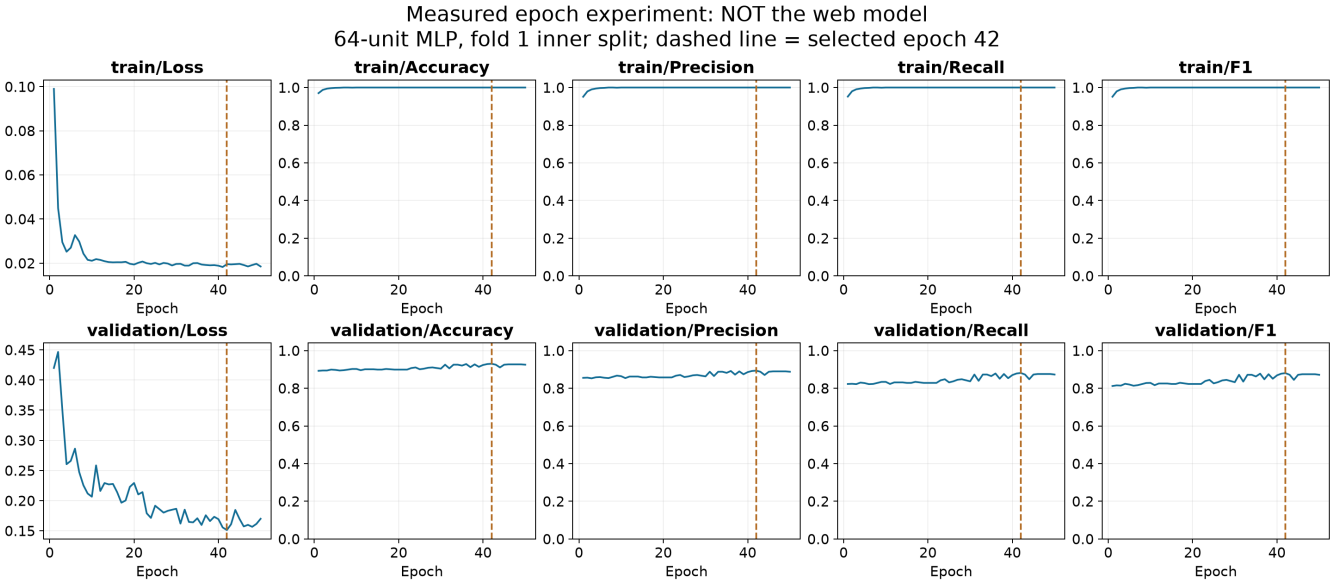
At 30 eggs per class, one error gives 96.67%; two give 93.33%. Confidence intervals still matter. Resample whole batches and keep repeated scans attached to their egg.

LIMIT / NEXT EVIDENCE

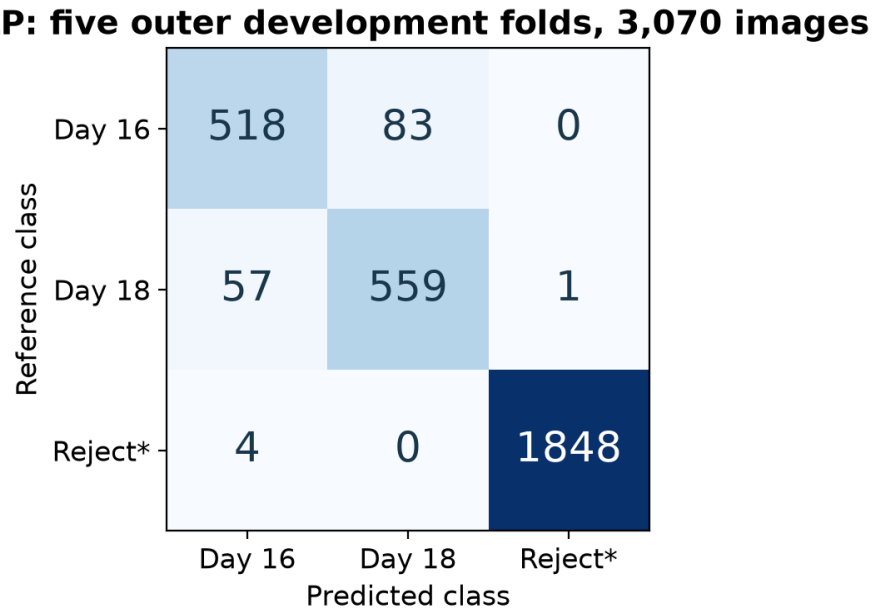
No new independently referenced eggs are available. Fresh physical score gap is unmeasured. A prepared form is not a completed field test.

EVIDENCE: scientific-field-protocol.txt; blank collection workbook; collect_scientific_field_scan.py and its validation tests.

Actual training graphs



Top row: training. Bottom row: validation. Loss should fall; higher accuracy, precision, recall and F1 are better. Gaps between training and validation can indicate limited generalization.



Curves: first inner fold, epochs 1-50. Precision, recall and F1 use macro averaging. Matrix: five outer development folds. These are different evaluation populations within the same MLP experiment.

Outer raw accuracy 95.28%. Physical-age recall: Day 16 88.33%, Day 18 92.78%. This candidate was not deployed.

EVIDENCE: Actual logged values in model-evidence.json; full-size PNGs beside this PDF. No invented YOLO loss or mAP graphs.

What improved, what did not

Selected development experiment	Raw overall	D16 recall	D18 recall
Grouped linear	95.31%	88.70%	94.63%
RBF full frame	96.03%	89.44%	95.56%
Interior representation	96.87%	95.19%	94.63%
Nested stacking	95.93%	89.81%	94.07%
Age hierarchy	96.03%	95.74%	95.00%
CPU MLP epochs	95.28%	88.33%	92.78%
Interior photometry	96.78%	95.00%	94.81%

236 configurations. Same 3,070 development images, 20 physical age eggs, five grouped folds. Age recalls use 540 physical views per age. Overall accuracy also includes 1,852 public Reject images.

Best selected overall result: 96.87%. Its physical-age recalls are 95.19% and 94.63%, both below 96%. Selection uses these development results, so they are not independent confirmation.

Same-fold linear control: Day 16 88.33%, Day 18 95.00%. The selected interior candidate improves Day 16 but slightly reduces Day 18. No complete promotion case is established.

SAY THIS

We measured improvements, but we did not replace the web model without meeting the full criteria. The next valid confirmation needs new independently labeled eggs.

Public evidence: model-evidence.json and PNGs beside this guide. Full local evidence: scientific packet, raw logs and scripts. Photos, accounts, credentials and private histories are not included here.